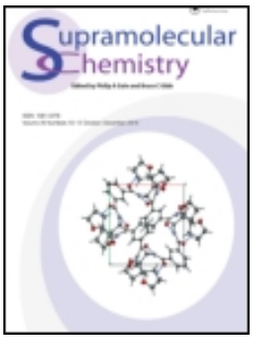

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Synthesis, structure, electrochemical behaviour and electrochemical investigations on the assembling with pyrene of a novel C_3 cryptand

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The synthesis and the structure of a new cryptand with 1,3,5-triphenylbenzene caps and pyridine rings in the chains are reported. The formation of a host–guest assembly between the investigated cryptand and pyrene was predicted by molecular modelling and confirmed by the electrochemical investigations by cyclic voltammetry using graphite electrodes.

Keywords: cryptand; pyrene-guest; molecular modelling; cyclic voltammetry

Introduction

The crown ethers, introduced by Pedersen (1967), exhibit a high affinity for alkaline and alkaline earth cations (*1*). Other similar compounds with nitrogen and/or sulfur instead of oxygen atoms in the macrocycles show a high complexation capacity for the cations of the transition (*d* block) metals (*1*). The selectivity of a given macrocycle for different guests is influenced by the heteroatoms belonging to the macrocycle, but also depends of the size and geometry of the cavity of the host system which must feet to the shape and size of the guest. As example, in crown ether series, the small [15]crown-5 has selectivity for Na^+, while the larger [18]crown-6 and [21]crown-7 reveal selectivity for K^+ and Cs^+, respectively (*1*). Due to their properties, crown ethers and similar macrocycles have widely been used for constructing highly selective ion sensors (*1,2*). Cryptands are even more efficient host systems showing higher ability and selectivity in the complexation processes (the macrobicyclic versus macrocyclic effect) (*1c*). Cryptands containing redox active or photochemical-sensitive groups exhibit interesting electron-transfer properties which can be explored for the development of highly sensitive and powerful electrochemical or photochemical methods for the investigation of host–guest interactions and molecular recognition (*3*). Sensors based on membrane electrodes for Hg^{2+}, Cd^{2+} and Pb^{2+} were constructed using the selective coordination ability for these cations of some cryptand type derivatives (*3c,d,f,h*).

Many cryptands and cyclophanes revealed high ability to form host–guest supramolecular system with aromatic hydrocarbons (e.g. anthracene, pyrene). The structure and the properties of these supramolecular architectures were investigated by single crystal X-ray diffractometry, NMR, UV–vis, MS spectra and electrochemical methods (CVs) (*4*). The recently published work of Stoddart and co-workers (*4*a) reports a plethora of host–guest supramolecules obtained with a cyclophane with extended paraquat units and various aromatic hydrocarbons. Cryptands with C_3 symmetry were considered of interest and many cryptands exhibiting 1,3,5-trisubstituted benzene (*5*), tertiary amines (*6*) or phosphines (*7*), cyclotribenzylene (*8*) and 1,3,5-triazine (*9*) units were successfully obtained and investigated.

In this context, the aim of this work is focused on the synthesis and the study of the properties of cryptand **1** (Chart 1) and the investigations of the possibility of the formation of supramolecular host–guest architectures between crytand **1** as host and pyrene as guest. The selection of pyrene as guest was supported by the results of preliminary molecular modelling investigations.

The formation of the pyrene ⊃ cryptand **1** complex was investigated by molecular modelling, NMR, UV–vis spectroscopies and cyclic voltammetry measurements (CV) using various supporting electrolytes.

Results and discussions

Compound **1** was obtained in fair yields (28%; Scheme 1) by the nucleophilic aromatic substitution reaction between triphenol **2** and 2,6-dichloro-3,5-dicyanopyridine **3** using a procedure adapted from similar reactions reported in the literature (*10*).

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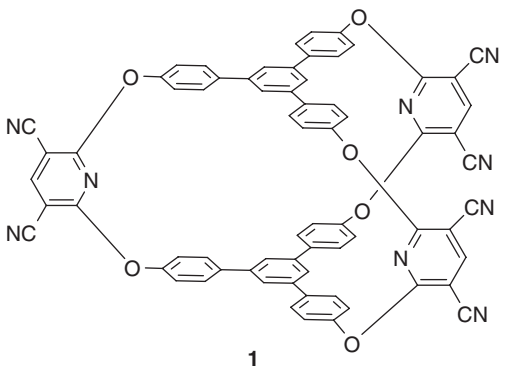


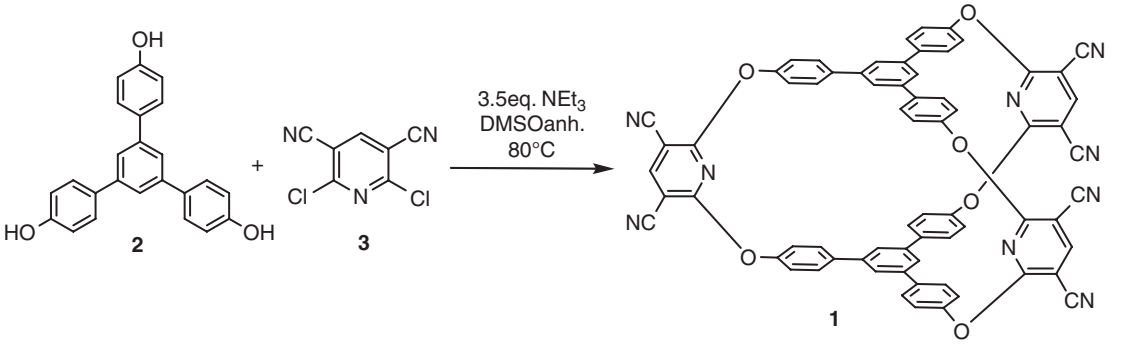
Chart 1. Target cryptand.

The theoretical models for the structure of the cryptand and of its complex with pyrene were obtained by *ab initio* calculations. The molecular modelling investigations started with the optimisation of the equilibrium geometry of the cryptand molecular structure. Its optimised structure is presented in Figure 1(A). The optimised geometry shows shifted parallel stacking structures of corresponding radial benzene molecules in the two stacking triphenylbenzene fragment. The central benzene rings show not only parallel stacking form but also have a coincident configuration in the two parallel triphenylbenzene plane. This configuration corresponds to the so-called 'sandwich' form of the benzene dimers (*11*). On the other hand, the presence of this sandwich configuration means that inside the single triphenylbenzene fragment radial benzene rings are rotated with about 120° (± 1°) compared to the central ring plane because of the charge repulsion between the hydrogen atoms of the central and the radial benzene rings. The intramolecular distance between the central benzene rings is 4.28 Å, which is larger than the equilibrium distance between benzene molecules (3.72 Å) in their sandwich configuration (*11*, *12*). The intermolecular distances between the planes of the shifted parallel stacking form of the radial benzene rings are around 3.7 Å, but the ring planes show small deviation from the ideal parallel configuration.

The shifted parallel stacking forms of the radial benzene rings have not exactly the same configuration as the equilibrium geometry of the stacked benzene dimers (*12*), where the benzene rings are shifted in the molecular plane with only 1.7 Å. In the present case, we have a larger shift of 2.41 Å, which is due to the same effect of the charge repulsion between the hydrogen atoms, discussed above.

The molecular cavity of the cryptand molecule can be illustrated as a polyhedron having the area of an equilateral triangle with side lengths of 12.46 Å and the height of 4.28 Å. The tilted positions of the radial benzene rings compared with the central rings configurations make somewhat difficult the intercalation of the guest pyrene molecule between the triphenylbenzene planes of the molecular cavity. Therefore, we have constructed an initial geometry of the host–guest system where we have already introduced the pyrene molecule between the planes of the triphenylbenzene. The equilibrium geometry of the cryptand–pyrene molecular complex is presented in Figure 1B. This geometry shows a distortion of the initial cryptand cavity, but also a binding structure of the pyrene inside the molecular cavity of the cryptand. The distorted triphenylbenzene fragments of the cryptand have curved structures, the side length of the equilateral triangle is reduced to the value of 12 Å and the distance between the central benzene rings is enlarged to 6.84 Å.

This distortion induces in the cryptand structure a deformation energy of 25.72 kcal/mol. On the other hand, this deformation energy is compensated by the intermolecular interaction energy of − 32.99 kcal/mol between the cryptand and pyrene, which gives the stability of the host–guest complex. Analysing the geometry of the complex, the pyrene molecular plane stays a little bit closer to the one of the planes of the central benzene rings (3.28 Å) as compared, for example, with the equilibrium inter-plane geometry of the graphene sheets (≈ 3.36 Å), while the distance to the other central benzene ring's plane is 3.35 Å. This closer packing configuration could generate an extra repulsion between the aromatic stacks which diminish a bit the magnitude of the "ideal" interaction



Scheme 1. Synthesis of cryptand **1**.

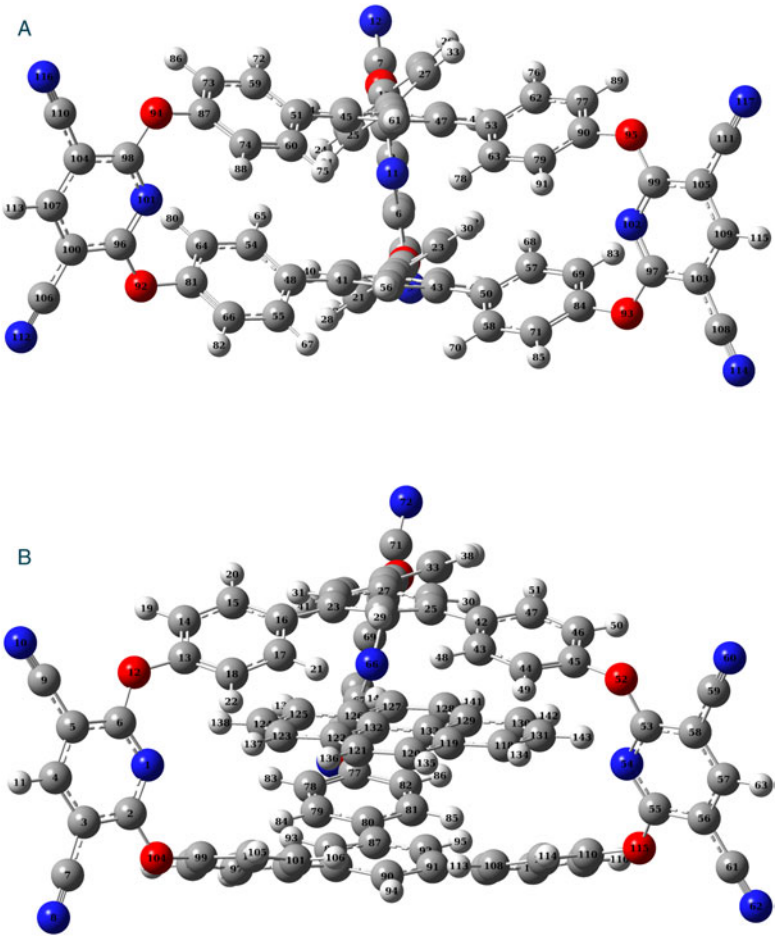


Figure 1. Representations of the optimised structures for cryptand **1** (A) and for its complex with pyrene (B).

energy between the aromatic sheets of the pyrene and the cryptand. This 'ideal' interaction energy can be obtained if the pyrene and the triphenylbenzene sheets would stay at their equilibrium distance.

The ^{1}H NMR spectra in solution (DMSO-d_6) revealed the flexibility of the chains and the signals correspond to an average symmetric structure. Thus, the protons of the central units and of the pyridine branches give singlets ($\delta = 7.66$ and 9.11 ppm, respectively), while the protons of the other aromatic rings give doublets ($\delta = 7.15$ and $\delta = 7.75$, $J = 8.4$ Hz). The ^{13}C NMR spectrum is also very simple exhibiting nine signals, one for the CN groups ($\delta = 113.00$ ppm), four signals for tertiary aromatic carbon atoms ($\delta = 121.26$; 127.06; 122.36 and 150.85 ppm) and five signals for the quaternary carbon atoms ($\delta = 89.14$; 136.84; 139.07; 150.23 and 164.00 ppm) (Figures 2(A) and (B)).

The investigations of the interactions of cryptand **1** with pyrene in solution started with NMR and UV–vis experiments. Surprisingly, the spectra of the mixtures of cryptand **1** with pyrene in different ratios (1:1; 1:2 and 1:3), even if the mixture was heated at 80°C for several hours (till 72 h) before the UV–vis and NMR measurements, did not exhibit relevant changes as compared to the spectra of cryptand **1** and pyrene themselves. We concluded that despite the thermodynamically favourable behaviour of the $1 \subset$ pyrene compared to the isolated molecules, the barrier for the entrance of the pyrene in the cavity of the cryptand is too high and the formation of the $1 \subset$ pyrene complex is not possible in the conditions of our experiments. The next step was the electrochemistry investigation of the formation of the $1 \supset$ pyrene, using the deposition of **1** on a graphite electrode.

The electrochemical behaviour of graphite electrode modified with cryptand **1** (G/cryptand electrode) was investigated comparatively with that of unmodified graphite electrode, by cyclic voltammetry in solution of pyrene, using different potential cycling (Figure 3). For unmodified graphite electrode, an anodic peak can be observed at ~ 65 mV versus Ag|AgCl/KCl$_{sat}$, this peak could be attributed to pyrene oxidation.

For G/cryptand electrode, the cyclic voltammograms present a well-defined peak pair, with the anodic peak potential of -12 mV versus Ag|AgCl/KCl$_{sat}$ and cathodic

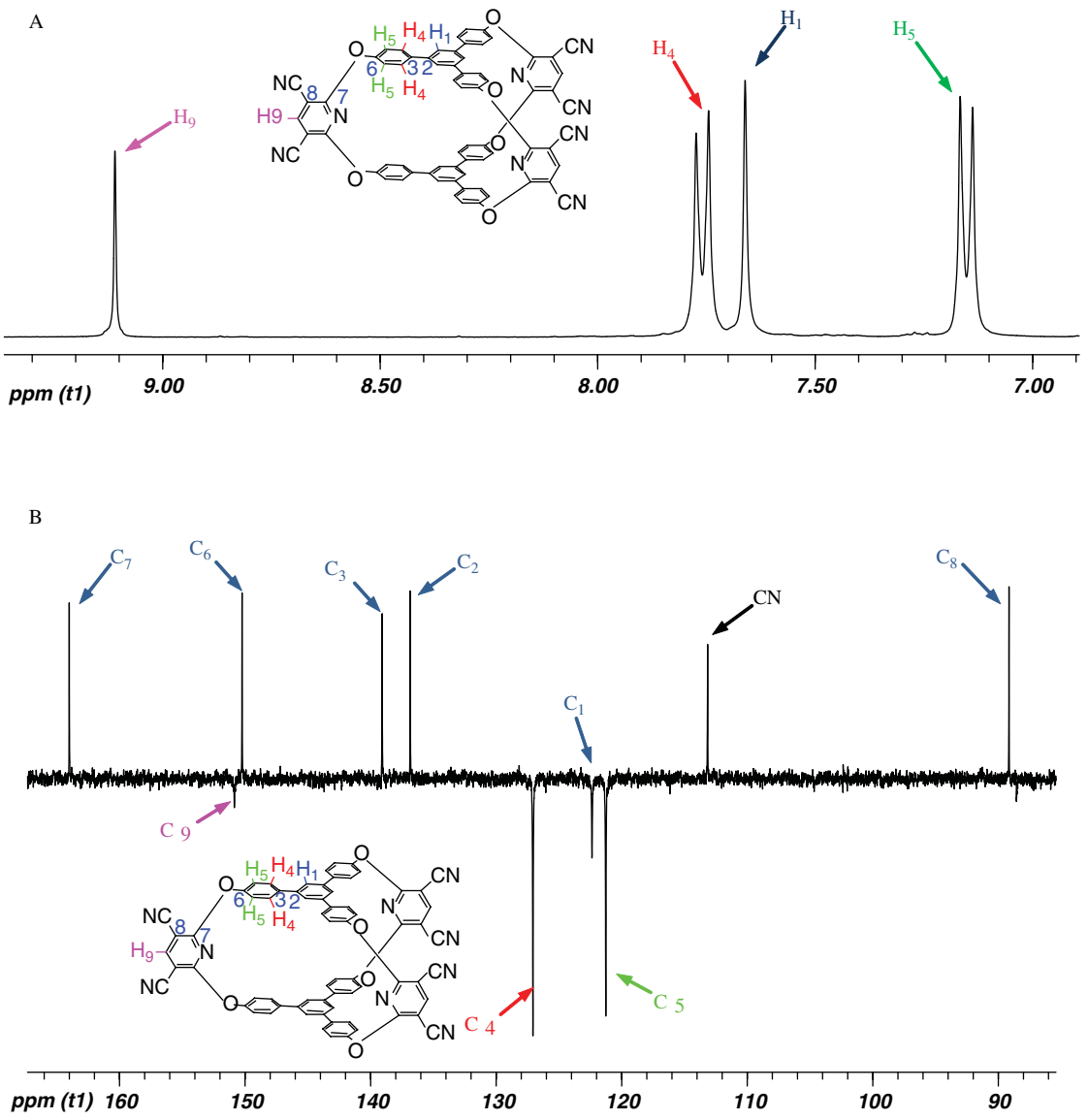


Figure 2. (Colour online) ^{1}H NMR (A) and ^{13}C NMR (B) spectra of cryptand **1**.

peak potential of −67 mV versus Ag|AgCl/KCl$_{sat}$, these peaks corresponding to cryptand oxidation and reduction. In a pyrene solution, at +144 mV versus Ag|AgCl/KCl$_{sat}$, the pyrene oxidation peak also appears. During cycling, it can be observed that all peak intensities decrease, the anodic pyrene potential moves towards negative potentials and this peak splits in two peaks (at 80 and 135 mV vs. Ag|AgCl/KCl$_{sat}$). These facts prove the formation of a new entity (compound) which is assigned to the formation of the cryptand ⊂ pyrene complex by the entrance of pyrene in the cryptand's cavity.

In Figure 4 are presented the cyclic voltammograms for the two electrodes: G/pyrene and G/cryptand + pyrene, in supporting electrolyte of phosphate buffer, pH 7. In the presence of cryptand, the same electrochemical behaviour is proved, with the first peak pair corresponding to cryptand oxidation/reduction at approximately the same potentials as in pyrene solution (−6 and respectively −72 mV vs. Ag|AgCl/KCl$_{sat}$) and the second oxidation peak of pyrene is also split but at different potentials (at 125 and 174 mV vs. Ag|AgCl/KCl$_{sat}$).

Based on Figure 4, the intensities of all four peaks were calculated during time cycling (20 cycles) and the linear regression parameters are presented in Table 1 together with the slope of peak current–cycling time dependence. It can be observed from Table 1 that for peak I and peak IV, the smallest intercepts were obtained and the slopes were positive, proving an activation process during cycling in phosphate buffer. Contrary to the electrochemical behaviour of peaks I and IV, for the other peaks

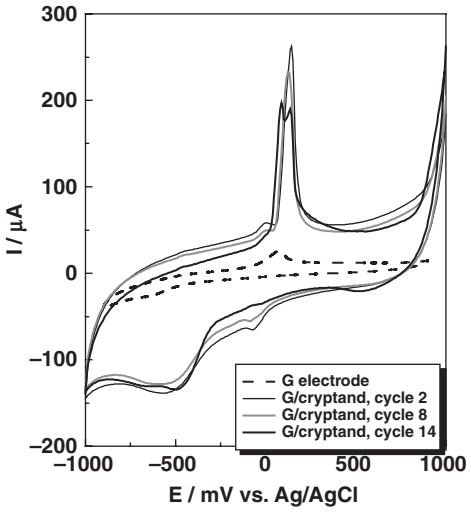


Figure 3. Cyclic voltammograms corresponding to graphite electrodes unmodified and modified with cryptand, during 14 cycles. Experimental conditions: starting potential, −1000 mV versus Ag|AgCl/KCl$_{sat}$; scan rate, 50 mV s^{-1}; supporting electrolyte, 5 mM pyrene solution.

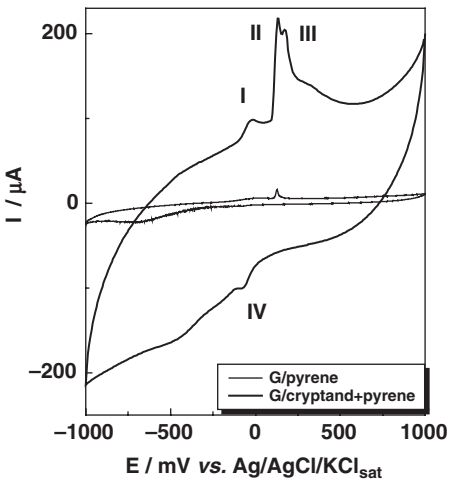


Figure 4. Cyclic voltammograms corresponding to graphite electrodes modified with pyrene; and cryptand and pyrene. Experimental conditions: starting potential, −1000 mV versus Ag|AgCl/KCl$_{sat}$; scan rate, 50 mV s^{-1}; supporting electrolyte, 0.1 M phosphate buffer, pH 7.

Table 1. Linear regression parameters corresponding to peak current dependence on cycling time for graphite electrodes modified with cryptand and pyrene.

Peak	$I_{t=0}$ (10^5 Å)	Slope (10^{10} Å s^{-1})	R/no. of exp. points
I	4.14	9.2	0.986/8
II	29.9	−10.9	0.962/7
III	24.1	−4.5	0.860/6
IV	5.03	7.1	0.964/9

Note: Experimental conditions: 0.1 M phosphate buffer, pH 7; for other conditions, see Figure 4.

corresponding to pyrene (II and III) a deactivation process took place.

This behaviour proved once again the pyrene entrance in the cryptand macrocycle, in both studies of electrochemical characteristics of cryptand and pyrene (cryptand first adsorbed on graphite electrode and supporting electrolyte was pyrene or cryptand and pyrene adsorbed on graphite electrode and supporting electrolyte was phosphate buffer, pH 7).

The results of the electrochemistry investigations and of the molecular modelling look to be in contradiction with the results obtained during the NMR and UV–vis experiments. However, the differences between the results of these experiments can be explained. Even if the **1** ⊂ pyrene system is more stable than the isolated molecules, in solution, the barrier for the penetration of pyrene in the cavity of the cryptand is too high and the host–guest system cannot be obtained even at 80°C. The obtaining of a positive result in the case of the electrochemical experiments which proved the formation of the **1** ⊂ pyrene complex can be explained by the experimental conditions used in cyclic voltammetry. The cryptand was first deposited on the graphite electrode. During this deposition, probably the conformation and the electronic behaviour of the cryptand are modified (e.g. the 1,3,5-triphenylene-benzene central unit(s) become(s) planar) and the barrier for the penetration of pyrene in the cavity of the cryptand is considerably diminished, and this process takes place during CV even at r. t. Modifications of the structure and electronic properties of proteins and aromatic compounds deposited (physisorbtion) on graphite or graphenes were largely observed and discussed (*13*).

Conclusions

The good yields synthesis of cryptand **1** was reported and the structure of the compound was confirmed by NMR and MS investigations. The formation of a cryptand ⊂ pyrene host–guest supramolecular architecture was supported by molecular modelling and electrochemical investigations.

Experimental part

General data

^{1}H NMR (300 MHz) and ^{13}C NMR (75 MHz) spectra were recorded in $CDCl_3$ at r.t. on 300 MHz using the solvent line as reference. Atmospheric pressure chemical ionisation (APCI, negative ions mode) were recorded on LTQ ORBITRAP XL spectrometer using external mass calibration. Melting point was determined with routine melting point apparatus and is uncorrected. Molecular modelling computations for equilibrium geometry structures and intermolecular interaction energy have been carried out at DFT level of theory using the M06-2X (*14*)

exchange–correlation functional and TZV triple-zeta valence basis set (*15*) implemented in the Gaussian09 program suite (*16*). M06-2X was parameterised to mimic short- and intermediate-range dispersion effects, and therefore M06-2X often gives reasonably good optimised geometry structure as well as intermolecular interaction energies for van der Waals complexes near equilibrium geometries, where the monomer densities overlap substantially (*17*, *18*).

Electrode preparation

A spectrographic graphite rod (Ringsdorff-Werke, GmbH, Bonn-Bad Godesberg, Germany) of ~3 mm diameter was wet polished on fine (grit 400 and 600) emery paper (Buehler, Lake Bluff, IL, USA). Then, a graphite rod of suitable length was carefully washed with deionised water, dried, and finally press-fitted into a PTFE holder in order to obtain an electrode having, in contact with the solution, a flat circular surface of ~0.071 cm^2.

The modified graphite electrodes were obtained by spreading onto the electrode surface 5 μL of 2 mM cryptand solution in acetonitrile, and leaving them for 24 h at room temperature to evaporate the solvent. Before immersion in the test solution, the modified electrodes were carefully washed with deionised water.

Electrochemical measurements

Electrochemical experiments were carried out using a typical three-electrode cell. The graphite electrode was used as working electrode, a platinum ring as counter electrode and an Ag|AgCl/KCl_{sat} as reference.

Cyclic voltammetry was performed on a PC-controlled electrochemical analyser (Autolab-PGSTAT 10, Eco-Chemie, Utrecht, The Netherlands).

The electrolyte solution was deoxygenated by bubbling argon for 15 min before measurements. The experimental results are the average of at least three identically studied electrodes, if not otherwise mentioned.

Synthesis

All reagents were of analytical grade and used without further purification. Thin layer chromatography (TLC) was conducted on silica gel 60 F254 TLC plates. Solvents were dried and distilled under argon using standard procedures. The supporting electrolyte was 0.1 M phosphate buffer solution, obtained by mixing solutions of $Na_2HPO_4{\cdot}2H_2O$ and $NaH_2PO_4{\cdot}H_2O$ in appropriate ratios. The pyrene and cryptand **1** solutions were obtained in dimethylsulfoxide (Merck).

Procedure: Triphenol **2** (0.10 g; 0.28 mmol), compound **3** (0.08 g; 0.42 mmol) and anhydrous NEt_3 (0.14 mL; 0.98 mmol) were solved at r.t., under Ar and under stirring in 10 mL DMSO. The temperature was raised till 80°C and the vigorous stirring was continued at this temperature over night (12 h). The reaction mixture was then cooled at r.t. and partitioned between ethylacetate (50 mL) and water (40 mL). The layers were separated and the aqueous layer was then extracted three times with ethylacetate (3 × 20 mL). The combined organic layers were washed with brine (50 mL), dried over Na_2SO_4, filtered and concentrated in vacuum. Pure compound **1** was obtained by crystallisation from CH_2Cl_2/EtOH = 1/1 as a white solid (yields 28%; 0.08 g).

Compounds **2** (*19*) and **3** (*20*) were already reported and they were obtained accordingly with the literature data.

10,12,30,32,47,49-hexaciano-8,14,28,34,45,51-hexaoxa-59,65,70-triaza-tridecacyclo [19.19.15.2.^{4,7}2.^{15,18}2.^{24,27}2.^{35,38}2.^{41,44}2.^{52,55}1.^{3,39}1.^{9,13}1.^{19,23}1.^{29,33}1.^{46,50}] doheptaconta-1,3(62),4,6,9(65),10,12,15,17,19,21,23(56),24,26,29(59),30,32,35,37,39,41,43,46(70), 47,49,52, 54,57,60,63,66,68,71-tritricontaene **1**.

White solid (yield 28%), m.p. >360°C. Calculated for $C_{69}H_{33}N_9O_6$: C, 64.45%; H, 3.07%; N, 11.63%. Found: C, 64.29%; H, 2.91%; N, 11.76%. ^{1}H NMR (300 MHz, DMSO-d_6) $\delta = 7.15$ (d, 12H, $J = 8.4$ Hz, 6-H, 16-H, 26-H, 36-H, 43-H, 53-H, 58-H, 60-H, 64-H, 66-H, 69-H, 71-H), 7.66 (s, 6H, 2-H, 20-H, 22-H, 40-H, 56-H, 62-H), 7.75 (d, 12H, $J = 8.4$ Hz, 5-H, 17-H, 25-H, 37-H, 42-H, 54-H, 57-H, 61-H, 63-H, 67-H, 68-H, 72-H), 9.11 ppm (s, 3H, 11-H, 31-H, 48-H). ^{13}C NMR (75 MHz, DMSO-d_6) $\delta = 89.14$ (10-C, 12-C, 30-C, 32-C, 47-C, 49-C), 113.00 (CN), 121.26 (6-C, 16-C, 26-C, 36-C, 43-C, 53-C, 58-C, 60-C, 64-C, 66-C, 69-C, 71-C), 122.36 (2-C, 20-C, 22-C, 40-C, 56-C, 62-C), 127.06 (5-C, 17-C, 25-C, 37-C, 42-C, 54-C, 57-C, 61-C, 63-C, 67-C, 68-C, 72-C), 136.84 (1-C, 3-C, 19-C, 21-C, 23-C, 39-C), 139.07 (4-C, 18-C, 24-C, 38-C, 41-C, 55-C), 150.23 (7-C, 15-C, 27-C, 35-C, 44-C, 52-C), 150.85 (11-C, 31-C, 48-C), 164.00 ppm (9-C, 13-C, 29-C, 33-C, 47-C, 49-C). MS (APCI−): $m/z = 1100.0$ $[M + OH]^-$, 1118.2 and 1120.1 $[M + Cl]^-$.

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Supplemental data

Supplemental data for this article can be accessed here: doi: 10.1080/10610278.2014.904868

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